LabaLab
Language, Affect and Behaviour in AI

ROBOT GUIDE

Furhat

Start it in the right order, find your way around Furhat Studio, run skills and build your own.



LabaLab · BGU · Room –132

MEET FURHAT

A face, ears, eyes and a voice

Face

Projected from inside onto a mask. Dozens of faces to choose from.

Ears

A four-microphone array in its base. Plug it in before you switch on.

Eyes

A camera that finds faces and turns Furhat’s gaze to the person talking.

Voice

A speaker inside, with hundreds of voices in many languages.

Everything runs on the robot. You control it from **Furhat Studio**, a web page the robot serves to your browser.

EVERY SESSION

Switch on, in this order

1. **WiFi on first**
   HOT-WiFi, or your hotspot on 2.4 GHz. Laptop on it too.
2. **Microphone plugged in**
   The mic array in the base, connected before power.
3. **Switch Furhat on**
4. **Wait for the face**
   Several minutes: it starts a whole computer. The face on the mask means ready.
5. **Open Studio and do the checks**
   Sign in (next slide), then the settings check.

⚠ Why the order matters

Furhat looks for WiFi, and loads its speech services from the cloud, **only while it starts**. It never tries again.

Start it before the WiFi and it wakes up **deaf and silent**, with no error.

The fix: WiFi on, then switch Furhat off and on.

FURHAT STUDIO

Open Studio, two ways

EASIEST

With Robot Hub

1. On the Furhat card, press **Connect**. Furhat says a sentence.

2. Press **Open Studio**. Robot Hub signs you in.

3. Card says **Needs password**? Press **Enter password**.

WITHOUT THE HUB

In any browser

1. Find Furhat’s address. It changes every session.

2. Type it into the browser’s address bar.

3. Sign in with the Studio password.

The Studio password is in the lab’s Bitwarden. Ask lab staff.

FURHAT STUDIO

Studio’s sections, one by one

Section	What it’s for
Home	Live camera with face tracking, a mic level meter and a Testing box. Your 30-second check.
Dashboard	The live conversation transcript, and the Wizard board to jump to any part of a script.
Skill Library	Start a built-in skill, or import your own **.skill** file.
Furhat AI Creator	Build a talking character without writing code.
Realtime API	Lets other programs drive Furhat over the network. Lab staff only.
Face Editor	Choose and adjust the face projected on the mask.
Settings	Voice, language, speaking, listening, audio devices and WiFi.

STUDIO · HOME · TESTING BOX

The 30-second check

Sight

Stand in front of Furhat. A green frame with an ID label appears around your face.

Hearing

Speak. The Microphone Levels bar moves. Flat: check Audio devices, or the mic was plugged in too late.

Speech

Speaking tab: Language English (United States), voice Ella22k_HQ. Type a sentence, press **Play**.

Face

Face tab: press a gesture like **BigSmile** or **Nod**. Look at the mask, not the screen.

Lips move but no sound? Do the Speaking and Listening flip on the next slide.

STUDIO · SETTINGS

Settings, and what to check

Settings ›	What to do
Speaking	**Every session:** each engine (Amazon Polly, Microsoft Azure) to Custom and straight back to Furhat provided.
Listening	Keep **Microsoft Azure**: Configured (Provided) in green. Never Google. Then the same Custom-and-back flip.
Audio devices	Pick **ReSpeaker 4 Mic Array**. It may start on Camera, which only hears up close.
Network	The current WiFi and address. **Add Network** for a new WiFi, then restart Furhat.
Face	Mask, character and projection. **Write the values down before changing: there is no undo.**

Why flip the engines? Furhat can show them as set while they never loaded: lips move, no sound.

STUDIO · SKILL LIBRARY

Skills: what Furhat does

A **skill** is a program that runs on Furhat: what it says, what it listens for, and how it reacts.

In the Skill Library

Built-in skills: ready to try, like MeetFurhat, JokeBot, Quiz and OpenAIChat.

Your skills: ones you imported as a .skill file.

Start: runs the skill now.

Autostart: runs it every time Furhat switches on.

Try one now

1. Open **Skill Library**.

2. Pick **JokeBot** and press **Start**.

3. Stand in front of Furhat and talk to it.

4. Open **Dashboard** to follow the conversation.

5. Back in Skill Library, stop it or start another.

THE LAB'S CHAT SKILL

OpenAIChat: open chat, faces, vision

1. **Open Robot files in Robot Hub**
   Settings › Lab staff tools › Robot files › Open. Ask lab staff.
2. **Pick Furhat, then OpenAIChat_1.3.0.skill**
3. **Press Download, With your GPT key**
   Your key is put inside your copy.
4. **Import it in Studio's Skill Library**
   Under Your skills.
5. **Press Start and talk**

Your copy has your key

The shared file has no key in it. Downloading **With** a key puts yours back.

Never pass your copy on. Share skills through Robot files instead: keys are stripped on upload.

STUDIO · DASHBOARD

Follow and steer a running skill

Live transcript

Everything Furhat says and hears, as it happens. If the person’s words don’t appear, Furhat isn’t hearing them: check listening and recognition.

Wizard board

Buttons for the parts of the skill’s script. Click one and Furhat jumps there. Handy for Wizard-of-Oz studies and for rescuing a demo.

CREATE A SKILL · NO CODE

Furhat AI Creator

1. **Open Furhat AI Creator in Studio**
2. **Describe the character in plain words**
   Who it is, how it behaves, what it talks about.
3. **Pick its face and voice**
   From Furhat’s faces and its hundreds of voices.
4. **Add knowledge, if it needs facts**
   Links to the information it should know.
5. **Talk to it, adjust, talk again**

Two ways to create

No code: AI Creator. A talking character in minutes, from a description.

With code: the Furhat SDK, in Kotlin. Full control of every line, gesture and turn. Next four slides.

AI Creator is part of Furhat’s paid FurhatAI plan. If it is locked, ask lab staff.

Start a new skill

1. **Open the Furhat SDK launcher**
   Press **Start Server** and wait until it says it is running.

2. **Press New Skill, give it a name**
   A popup shows where it was saved. Copy that path.

3. **Open the folder in VS Code**

```
cd <the path>
code .
```

4. **Edit flow/interaction.kt**
   The only file you change.

You need

The Furhat SDK launcher (from furhat.io, with a developer account).

VS Code.

Java JDK 21, with JAVA_HOME set, to build.

Tip: paste interaction.kt into an AI assistant, describe what Furhat should do, paste the result back.

Written in Kotlin, laid out like this

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Kotlin

Every Furhat skill is written in **Kotlin**, with the Furhat SDK. Java libraries work too.

You don’t need to be a Kotlin expert: the SDK gives you short commands like say, ask and goto.

In the lab you edit only **flow/interaction.kt**, in VS Code.

File or folder	What goes in it
main.kt	The starting point. It starts the flow.
flow/interaction.kt	Your conversation, from its Start state.
flow/general.kt	Waiting and noticing people.
nlu/nlu.kt	Intents: what people might say.
users.kt	What to remember about each person.
assets/	Extra files, like a screen page.
build.gradle	The SDK version and libraries.

A skill is a set of states

Each **state** is one moment of the conversation. It reacts to what happens, then moves on:

onEntry · when the state starts

onResponse · the person said something that matches

onNoResponse · the person stayed silent

onUserEnter / onUserLeave · someone arrives or leaves

onButton · adds a button to the Wizard board

goto · move to the next state

EXAMPLE

```
val Greeting = state {
  onEntry {
    furhat.ask("Want to play a game?")
  }
  onResponse<Yes> {
    furhat.say("Great, let's start!")
    goto(Game)
  }
  onResponse<No> {
    furhat.say("Maybe another time.")
    goto(Goodbye)
  }
}
```

CREATE A SKILL · WITH CODE · 4 OF 4

Build, test, put it on the robot

BUILD AND TEST ON THE LAPTOP

1. In VS Code: Terminal › New Terminal, then run:

```
.\gradlew shadowJar
```

2. Wait for **BUILD SUCCESSFUL**.

3. In the launcher: **Start Skill** › build › libs › your **.skill** file.

4. Test it at **localhost:8080**, on the virtual Furhat.

ON THE REAL ROBOT

5. Open Furhat Studio on the robot.

6. Skill Library › Your Skills: import the .skill file.

7. Press **Start**.

EVERY CHANGE

Edit, build again, then stop and start the skill with the new file.

A key used in a skill is built **into** the .skill file. Never share a built skill: share through Robot files.

WHEN SOMETHING GOES WRONG

What you see, and what to do

What you see	What to do
Lips move, no sound	The engines didn’t load. Settings › Speaking: Custom, then back to Furhat provided.
It can’t hear you	Recognition must be Microsoft Azure. Check the mic meter on Home.
Mic meter stays flat	The mic was plugged in after power-on. Restart Furhat with it plugged in.
English in a Japanese voice	Set the language to English (United States) and pick an English voice.
Fine yesterday, silent today	It started before the WiFi. WiFi on, then switch Furhat off and on.
Not on the network	Hotspot on 2.4 GHz, and on before Furhat starts. Then restart Furhat.
Card says Needs password	Press Enter password, or Robot Hub › Settings › Furhat password.

REMEMBER

Four habits for a smooth session

WiFi and mic first

Both ready before you switch Furhat on.

Settings check, every time

Flip the engines, check Azure and the language.

Test on Home first

Camera, mic, speech, gestures: 30 seconds before people arrive.

Keep keys private

A skill downloaded with your key stays with you.